

# BSDS V: Regression Techniques

## Problems Set 1

### 1 Random Vectors

1. Show the following equalities:

(a)  $\text{Var}(\underset{\sim}{Z} - \underset{\sim}{\beta}) = \text{Var}(\underset{\sim}{Z}).$

(b)  $\text{Cov}(\underset{\sim}{a}^\top \underset{\sim}{Z}, \underset{\sim}{b}^\top \underset{\sim}{W}) = \underset{\sim}{a}^\top \text{Cov}(\underset{\sim}{Z}, \underset{\sim}{W}) \underset{\sim}{b}.$

(c)  $\text{Cov}(\underset{\sim}{X}_1 + \underset{\sim}{X}_2, \underset{\sim}{Z}_1 + \underset{\sim}{Z}_2) = \text{Cov}(\underset{\sim}{X}_1, \underset{\sim}{Z}_1) + \text{Cov}(\underset{\sim}{X}_1, \underset{\sim}{Z}_2) + \text{Cov}(\underset{\sim}{X}_2, \underset{\sim}{Z}_1) + \text{Cov}(\underset{\sim}{X}_2, \underset{\sim}{Z}_2).$

(d)  $\text{Var}(\underset{\sim}{X} \pm \underset{\sim}{Z}) = \text{Var}(\underset{\sim}{X}) + \text{Var}(\underset{\sim}{Z}) \pm \text{Cov}(\underset{\sim}{X}, \underset{\sim}{Z}) \pm \text{Cov}(\underset{\sim}{Z}, \underset{\sim}{X}).$

(e)  $E(\underset{\sim}{Z}^\top A \underset{\sim}{Z}) = \underset{\sim}{\mu}^\top A \underset{\sim}{\mu} + \text{tr}(A \Sigma),$  where  $E(\underset{\sim}{Z}) = \underset{\sim}{\mu}, \text{Var}(\underset{\sim}{Z}) = \Sigma.$

2. Let  $X = ((X_{i,j}))$  be an  $m \times n$  matrix. Define  $E(X) = ((E(X_{i,j})))$ . Let  $A = ((a_{i,j}))$  and  $B = ((b_{i,j}))$  be non-random matrices. Then show that  $E(AXB) = AE(X)B$ .

3. Let  $X_1, \dots, X_n$  be independent random variables with  $E(X_i) = 0, \text{var}(X_i) = \mu_2, E(X_i^3) = 0$  and  $E(X_i^4) = \mu_4$  for all  $i = 1, \dots, n$ . Let  $A = ((a_{i,j}))$  be a non-random symmetric matrix. Then show that

$$\text{var}(\underset{\sim}{X}^T A \underset{\sim}{X}) = \mu_2^2 \left( 2 \sum_{i,j} a_{i,j}^2 - 3 \sum_i a_{i,i}^2 \right) + \mu_4 \left( \sum_i a_{i,i}^2 \right).$$

### 2 Differentiation

1. Find the maxima and/or minima of the following functions:

(a)  $g(\underset{\sim}{x}) = \|\underset{\sim}{y} - A\underset{\sim}{x}\|^2$  where  $A$  is a p.d. invertible matrix.

(b)  $g(\underset{\sim}{x}) = \sum_{j=1}^k (x_j^4 - 4x_j^2)$

(c)  $g(\underset{\sim}{x}) = \|\underset{\sim}{x}\|^2 + \exp\{\underset{\sim}{q}^T \underset{\sim}{x}\}$

2. Let  $(X, Y)$  have the joint pdf

$$f_{X,Y}(x, y) = \begin{cases} 2, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Define the transformation  $U = X + Y, V = X - Y$ . Find the joint distribution of  $(U, V)$ .

3. Let  $(X, Y)$  have the joint pdf

$$f_{X,Y}(x, y) = \begin{cases} 4xy, & 0 < x < 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Define the transformation  $U = XY, V = Y$ . Find the joint distribution of  $(U, V)$ .

4. Let  $(X, Y)$  have the joint pdf

$$f_{X,Y}(x, y) = \begin{cases} e^{-(x+y)}, & x > 0, y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Define the transformation  $U = X$ ,  $V = Y/X$ . Find the joint distribution of  $(U, V)$ .

5. Let  $(X_1, X_2, \dots, X_k)$  have the joint pdf

$$f_{X_1, \dots, X_k}(x_1, \dots, x_k) = \begin{cases} e^{-(x_1 + x_2 + \dots + x_k)}, & x_i > 0, i = 1, \dots, k, \\ 0, & \text{otherwise.} \end{cases}$$

Define the transformation

$$U_1 = X_1, \quad U_i = \frac{X_i}{X_1}, \quad i = 2, \dots, k.$$

Find the joint distribution of  $\{U_1, \dots, U_n\}$ .

### 3 Multivariate Normal Distribution

1. Let  $X \sim \text{normal}(0, 1)$ . Define  $Y = \text{sign}(X)$  and  $Z = |X|$ . Here  $\text{sign}(\cdot)$  is a  $\mathbb{R} \rightarrow \{0, 1\}$  function such that  $\text{sign}(a) = 1$  if  $a \geq 0$ , and  $\text{sign}(a) = -1$  otherwise.

- Find the marginal distributions of  $Y$  and  $Z$ .
- Find the joint CDF of  $(Y, Z)$ . Hence or otherwise prove that  $Y$  and  $Z$  are independently distributed.

2. Let the joint distribution of  $\underline{X}$  be an equal mixture of  $N_4(\underline{0}, \Sigma_1)$  and  $N_4(\underline{0}, \Sigma_2)$ , where  $\Sigma_1 = \begin{bmatrix} E_{2,\rho} & 0 \\ 0 & E_{2,\rho} \end{bmatrix}$  and  $\Sigma_2 = \begin{bmatrix} E_{2,\rho} & 0 \\ 0 & E_{2,-\rho} \end{bmatrix}$ , and  $E_{2,\rho}$  is the  $2 \times 2$  equi-correlation matrix with  $-1 < \rho < 1$ .

- Find the joint pdfs of  $(X_1, X_2)$  and  $(X_3, X_4)$ .
- Prove that  $X_4$  and  $X_3$  are uncorrelated but not independent, and
- Are  $(X_1, X_2)$  and  $(X_3, X_4)$  independent?

3. Let  $\underline{X} = (X_1, X_2, X_3)^\top$  be a random vector with joint pdf

$$f_{\underline{X}}(\underline{x}) = (2\pi)^{-3/2} \exp\{-(x_1^2 + x_2^2 + x_3^2)/2\} \times \{1 + x_1 x_2 x_3 \exp\{-(x_1^2 + x_2^2 + x_3^2)/2\}\}, \quad x_i \in \mathbb{R}, \forall i = 1, 2, 3.$$

- Show that  $X_i \sim N(0, 1)$  for all  $i = 1, 2, 3$ .
- Show that  $(X_i, X_j)$  jointly follow a bivariate normal distribution. Hence find the expectation and variance-covariance matrix of  $\underline{X}$ .
- Does  $\underline{X}$  follow a multivariate normal distribution. Justify your answer.